Homework 1 (Due September 24 before class)

Grade Distribution (Total=10+10+20+15+10+15+10=90)

1.

Exercise 11.1 (Bayes Formula)

(1) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, and let $(A_1, A_2, \ldots, A_n)$ be a measurable partition of Ω such that $\mathbb{P}(A_i) > 0$ for every $i \in \{1, \ldots, n\}$. Prove that, for every $B \in \mathcal{A}$ such that $\mathbb{P}(B) > 0$, for every $i \in \{1, \ldots, n\}$,

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B \mid A_i)}{\sum_{j=1}^{n} \mathbb{P}(A_j)\mathbb{P}(B \mid A_j)}.$$

(2) Suppose that we have n boxes numbered $1, 2, \ldots, n$, and that the i-th box contains r_i red balls and n_i black boxes, where $r_i, n_i \geq 1$. Imagine that one chooses a box uniformly at random, and then picks a ball (again at random) in the chosen box. Compute the probability that the i-th box was chosen knowing that a red box was picked.

2.

Exercise 5.1.2. Bayes' formula. Let $G \in \mathcal{G}$ and show that

$$P(G|A) = \int_G P(A|\mathcal{G})\, dP \Big/ \int_\Omega P(A|\mathcal{G})\, dP$$

When $\mathcal{G}$ is the σ-field generated by a partition, this reduces to the usual Bayes' formula

$$P(G_i|A) = P(A|G_i)P(G_i) \Big/ \sum_j P(A|G_j)P(G_j)$$

3.

Exercise 11.8 Let $(\mathcal{A}_n)_{n\in\mathbb{N}}$ be a decreasing sequence of sub-σ-fields of $\mathcal{A}$, with $\mathcal{A}_1 = \mathcal{A}$, and let $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$.

(1) Prove that the random variables $\mathbb{E}[X \mid \mathcal{A}_n] - \mathbb{E}[X \mid \mathcal{A}_{n+1}]$, for $n \in \mathbb{N}$, are orthogonal in L^2, and that the series

$$\sum_{n\in\mathbb{N}} \big(\mathbb{E}[X \mid \mathcal{A}_n] - \mathbb{E}[X \mid \mathcal{A}_{n+1}]\big)$$

converges in L^2.

(2) Let $\mathcal{A}_\infty = \cap_{n\in\mathbb{N}} \mathcal{A}_n$. Prove that

$$\lim_{n\to\infty} \mathbb{E}[X \mid \mathcal{A}_n] = \mathbb{E}[X \mid \mathcal{A}_\infty], \quad \text{in } L^2.$$

4.

Exercise 5.1.7. Show that when $E|X|$, $E|Y|$, and $E|XY|$ are finite, each statement implies the next one, and give examples with $X, Y \in \{-1, 0, 1\}$ a.s. that show the reverse implications are false: (i) X and Y are independent, (ii) $E(Y|X) = EY$, (iii) $E(XY) = EXEY$.

5.

Exercise 5.1.11. Show that if X and Y are random variables with $E(Y|\mathcal{G}) = X$ and $EY^2 = EX^2 < \infty$, then $X = Y$ a.s.

6.

Exercise 5.1.12. The result in the last exercise implies that if $EY^2 < \infty$ and $E(Y|\mathcal{G})$ has the same distribution as Y, then $E(Y|\mathcal{G}) = Y$ a.s. Prove this under the assumption $E|Y| < \infty$. Hint: The trick is to prove that $\operatorname{sgn}(X) = \operatorname{sgn}(E(X|\mathcal{G}))$ a.s., and then take $X = Y - c$ to get the desired result.

7. Prove the conditional Fatou's lemma: If $X_1, X_2, \cdots$ are nonnegative random variables and $\mathcal{F}$ is a σ-field, then

$$E\left[\liminf_{n\to\infty} X_n \middle| \mathcal{F}\right] \le \liminf_{n\to\infty} E[X_n|\mathcal{F}].$$

Hint: Check the proof for the usual Fatou's lemma (see, e.g., Theorem 1.5.4 of Durrett).